

Kwanwoo Lee

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EDUCATION

Seoul National University

Integrated M.S./Ph.D. Program in Intelligence and Information; GPA: 4.00/4.5

Inha University

Bachelor of Science in Mechanical Engineering; GPA: 4.42/4.5

- Summa Cum Laude; ranked 1st out of 59 students

Seoul, Republic of Korea

Aug. 2022 – Present

Incheon, Republic of Korea

Mar. 2017 – Aug. 2022

RESEARCH INTERESTS

Humanoid Robotics: Humanoid locomotion, whole body control, loco-manipulation

Motion Retargeting: Collision aware motion retargeting, dynamically feasible motion optimization, human-to-humanoid motion transfer (**Video:** <https://youtu.be/ruav1TcfNyw>)

Safety Critical Control: Control barrier functions, Constrained reinforcement learning, Safe reinforcement learning

Robot Learning: Reinforcement learning for humanoid locomotion, general motion tracking, imitation learning

PROJECTS

민관협력 기반 AI 휴머노이드 원천기술 고도화 | 과학기술정보통신부

2026

- Developed a safety-aware command filtering framework for humanoid robots to enable dynamically balanced locomotion and manipulation.
- **Demo:** [youtube.com/watch?v=CL3e6qg7Mxk](https://youtu.be/CL3e6qg7Mxk)
- Designed motion retargeting methods that map human motion data to humanoid robots while considering kinematic and dynamic feasibility.
- Utilized retargeted humanoid motions as high-quality training data for reinforcement learning-based whole-body control policies.
- **Collision Aware Motion Retargeting:** <https://youtu.be/ruav1TcfNyw>

AI 파운데이션 모델 기반 가정 및 의료 환경 특화 휴머노이드 로봇 개발 | 산업통상자원부

2025 – 2026

- Developed RL-based whole-body controllers for BlueRobin's humanoid robot platform.
- Implemented humanoid locomotion and manipulation controllers, robot physics simulation environments, motor communication modules, and state estimation pipelines.
- Participated in CES 2026 demonstration and technical preparation for humanoid robot deployment.
- **Media:** CES 2026 coverage
- **Demo:** [youtube.com/watch?v=VUrm83OZvyA](https://youtu.be/VUrm83OZvyA)

다양한 응용 태스크를 수행할 수 있는 휴머노이드 AI 로봇 파운데이션 모델 국산화 | 산업통상자원부

2025 – 2026

- Developed RL-based locomotion controllers for humanoid robots.
- **Demo:** [youtube.com/watch?v=8Zdxw3Zo_Ds](https://youtu.be/8Zdxw3Zo_Ds)

내재 센서 기반 전신보행 제어 및 실시간 동작 계획 | 과학기술정보통신부

2022 – 2024

- Developed a model-based control framework for robust balance stabilization of humanoid robots during walking.
- **Demo:** [youtube.com/watch?v=8drBIHnn3zg](https://youtu.be/8drBIHnn3zg)

모바일 로봇 시뮬레이션을 위한 물리엔진 평가 | 삼성 SAIT AMD

2022 – 2023

- Evaluated physics engines and robot simulators for accurate and efficient mobile robot simulation.
- **Publication:** Journal of Korea Robotics Society paper

상지 움직임 기반 인간 보행주기 추정 | 하이젠알앤엠

2022 – 2023

- Developed a gait phase estimation method using human upper body motion.

PUBLICATIONS

- [1] **K. Lee**, S. Park, G. Park, M.-J. Kim, and J. Park, “Safety-Critical Whole-Body Control for Humanoid Robots via Input-to-State Safe Control Barrier Functions,” *arXiv preprint arXiv:2605.25546*, 2026.
- [2] M.-J. Kim, D. Lim, G. Park, **K. Lee**, and J. Park, “A Model Predictive Capture Point Control Framework for Robust Humanoid Balancing via Ankle, Hip, and Stepping Strategies,” *IEEE Transactions on Robotics*, 2025.
- [3] G. Park, M.-J. Kim, **K. Lee**, and J. Park, “Humanoid Walking Stabilization via Model Predictive Control with Step Adjustment Based on the 3D Divergent Component of Motion,” in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pp. 14707–14713, 2025.
- [4] J. Kim, **K. Lee**, S. Park, and J. Park, “Real-Time Whole-Body Model Predictive Control for Versatile 3D Bipedal Locomotion via Novel Kino-Dynamic Model,” *IEEE Access*, 2025.
- [5] **K. Lee**, G. Park, M.-J. Kim, and J. Park, “Phase-based Nonlinear Model Predictive Control for Humanoid Walking Stabilization with Single and Double Support Time Adjustments,” *arXiv preprint arXiv:2506.03856*, 2025.
- [6] S. Park, **K. Lee**, and J. Park, “Balance Control of Humanoid Robots Using Control Barrier Functions for Self-Collision Avoidance and Centroidal Angular Momentum,” in *Proceedings of the Institute of Control, Robotics and Systems Conference*, pp. 259–260, 2025.
- [7] M.-J. Kim, D. Lim, D. Kim, J. Cha, J. Shin, W. Cha, G. Park, **K. Lee**, and J. Park, “Advances in Humanoid Robot Walking Technologies: A Review,” *Journal of Institute of Control, Robotics and Systems*, vol. 30, no. 4, pp. 412–422, 2024.
- [8] **K. Lee**, J. Yoon, S. Park, and J. Park, “Performance Evaluation of Robotic Physics Engine for Mobile Manipulator Simulation,” *The Journal of Korea Robotics Society*, vol. 19, no. 1, pp. 31–38, 2024.

TEACHING

Head Teaching Assistant <i>Seoul National University</i>	2025 – 2026
• Theory and Practice of Humanoid Walking Control, Spring 2025; Instructor: Prof. Jaeheung Park. • Theory and Practice of Humanoid Walking Control, Spring 2026; Instructor: Prof. Jaeheung Park.	
Teaching Assistant <i>Seoul National University</i>	2022 – 2025
• DYROS Robotics Boot Camp I: ROS, Robot Operating System; Instructor: Prof. Jaeheung Park.	

TECHNICAL SKILLS

Programming Languages: Python, C/C++, MATLAB
Robotics & Control: Humanoid locomotion, whole-body control, model predictive control, control barrier functions, inverse kinematics, state estimation, humanoid motion retargeting
Robot Software: ROS, ROS2, MuJoCo, Pinocchio, Crocoddyl, IsaacLab, IsaacSim, RViz
Robot Learning: Reinforcement learning, imitation learning, sim-to-real transfer, actuator network modeling
Optimization & Numerical Tools: Quadratic programming, nonlinear optimization, CasADi, qpOASES, qpSWIFT
Developer Tools: Git, Linux, CMake, VSCode

LANGUAGES

Korean: Native
English: Intermediate; TEPS 341